Adaptive Hazard Intelligence:

A Calibrated Multi-Hazard Risk Engine for County Emergency Management

BAS-EM Capstone Project

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Abstract

Emergency managers need calibrated inputs to make defensible resource allocation decisions, not dashboards that show hazards without quantifying confidence. Commercial catastrophe models are expensive, opaque, and optimized for insurance portfolios rather than public safety. Preparedness tools identify hazards but provide no basis for prioritization. Adaptive Hazard Intelligence (AHI) occupies the operational middle ground: a calibrated risk engine that returns actionable confidence tiers so practitioners retain control over escalation decisions.

AHI integrates federal datasets from NOAA, USGS, Census, CDC, FEMA, and DHS through Hazard-LM, a transformer architecture employing diffusion-based (heat kernel) attention that simultaneously predicts wildfire, flood, wind, winter storm, and seismic risks at the county level. Trained on 370,000 county-day observations from Washington State spanning 2000 to 2025, the system uses a multi-stage calibration pipeline—temperature scaling, seasonal priors, and base-rate ceilings—to produce probabilities aligned with historical event frequencies. An XGBoost baseline achieves strong discrimination (mean AUC 0.78; fire 0.87, winter 0.89) with low calibration error (ECE 0.004–0.054). The diffusion model achieves AUC 0.73 for fire and 0.74 for winter, with weaker performance on wind (0.59) and seismic (0.50) hazards—an honest reflection of the difficulty of predicting rare, stochastic events from weather features alone. A post-hoc calibration pipeline brings deployed predictions into alignment with base rates across all hazard types. The system runs on commodity hardware, making calibrated hazard intelligence accessible to under-resourced jurisdictions. A working deployment is available on Streamlit Cloud with full source code on GitHub.

1 Executive Summary

Adaptive Hazard Intelligence (AHI) gives emergency managers what they actually need: calibrated risk estimates with clear confidence tiers that support defensible resource allocation decisions.

Existing tools fall into two camps that both miss the mark:

- **Commercial catastrophe models**: Expensive, opaque, insurance-centric, and unable to explain their outputs to practitioners
- **Preparedness tools**: Identify hazards but provide no basis for prioritization or escalation decisions

AHI occupies the operational middle ground with six key differentiators:

1.1 Key Differentiators

- **Multi-hazard prediction**: Five hazard types in one calibrated system (rare in the field)
- **County-scale resolution**: The operational sweet spot for emergency management
- **Calibration over discrimination**: Post-hoc calibration pipeline ensures probabilities match historical frequencies
- **Confidence tiers**: Clear operational guidance on when to act, verify, or monitor
- **Practitioner control**: Quantified confidence enables informed escalation decisions
- **Accessible deployment**: Runs on commodity hardware, deployed to Streamlit Cloud

1.2 Operational Metrics

- Training data: 370,383 county-day observations across 39 Washington counties (2000 to 2025)
- XGBoost baseline: Mean AUC 0.78 (fire 0.87, winter 0.89, seismic 0.72, wind 0.71, flood 0.71)
- HazardLM-Diffusion: Mean AUC 0.64 (fire 0.73, winter 0.74, flood 0.65, wind 0.59, seismic 0.50)
- Calibrated deployment: Temperature scaling + seasonal priors + base-rate ceilings
- Model size: 880,000 parameters (3-layer transformer, 128-dim hidden)
- Deployment: Live on Streamlit Cloud, source on GitHub

2 Introduction

2.1 The Gap: Hazard Awareness Without Decision Support

Emergency managers have access to extensive hazard information but limited tools for translating that information into prioritized action. The core operational questions are:

1. *How should I prioritize* resources given current conditions across multiple hazard types?
2. *What confidence level* supports this prioritization? Should I act now, verify locally, or continue monitoring?
3. *Can I defend* this resource allocation decision to elected officials and the public?

Current tools address these questions poorly. Large language models generate text about disasters but cannot quantify risk. Commercial catastrophe models provide probabilistic outputs but are expensive, insurance-centric, and opaque to practitioners. Preparedness tools catalog vulnerabilities but provide no basis for escalation decisions.

2.2 Research Objectives

This project develops Adaptive Hazard Intelligence (AHI), a calibrated risk engine that addresses these gaps:

1. **Calibrated Prediction**: Probabilities that reflect true event frequencies, achieved through a multi-stage post-hoc calibration pipeline
2. **Operational Confidence Tiers**: Clear guidance on when to act, verify with local observations, or continue routine monitoring
3. **Accessible Deployment**: Runs on commodity hardware, deployed to Streamlit Cloud, with full source code available for audit and adaptation

2.3 Relationship to Prior Work

Where prior research has focused on improving how emergency managers communicate risk to the public [1], AHI focuses on improving the quality and interpretability of the risk estimates themselves. These efforts are complementary: better calibrated inputs enable clearer, more defensible communications.

3 System Architecture

AHI consists of two integrated subsystems: a data integration platform that harmonizes federal datasets, and the Hazard-LM prediction engine that produces calibrated probabilities. An optional LLM summarization layer translates outputs for practitioner-facing interfaces.

3.1 Data Integration Platform

3.1.1 Federal Data Sources

The platform integrates data from multiple federal agencies to provide comprehensive hazard context:

Table 1: Integrated Federal Datasets

Source	Dataset	Records	Update Freq.
NOAA	GridMET Weather	227,916	Daily
NOAA	Storm Events	6,124	Monthly
Census	Population Estimates	4,446	Annual
CDC	Social Vulnerability Index	158 vars	Biennial
FEMA	Disaster Declarations	68,232	Continuous
DHS	Critical Infrastructure	52,057	Quarterly
USGS	Earthquake Catalog	16,527	Continuous
USGS	Streamflow Gauges	26,789	Hourly

3.1.2 Data Processing Pipeline

Raw data undergoes standardization through spatial alignment (county-level aggregation using centroid-based assignment), temporal alignment (daily resolution with 14-day lookback windows), Z-score normalization relative to historical baselines, and automated quality control with missing value imputation.

3.2 HazardLM-Diffusion Architecture

HazardLM-Diffusion is a compact transformer (880,000 parameters) that processes tabular and temporal inputs through specialized encoders before producing multi-hazard predictions via per-hazard prediction heads with Low-Rank Adaptation (LoRA).

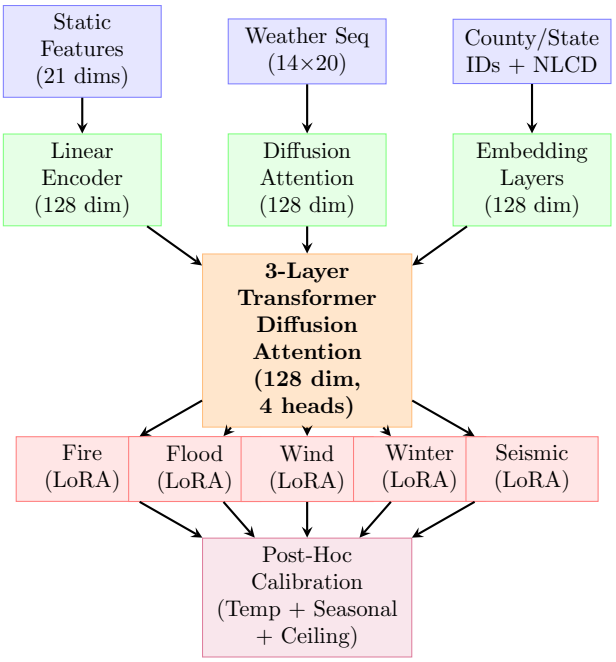

Figure 1: HazardLM-Diffusion v2.0 Architecture

3.2.1 Input Modalities

Static Features (21 dimensions) County-level attributes including geography (latitude, longitude, elevation), date encoding (day of year, month, year), meteorological observations (maximum and minimum temperature, relative humidity, wind speed, Energy Release Component, precipitation, vapor pressure deficit), derived indicators (3-day rolling means for temperature, precipitation, and wind; red flag fire weather warnings), and land characteristics (forest fraction, urban fraction, population density).

Temporal Sequence (14 days $\times$ 20 features) Daily weather observations forming a time series, zero-padded in the current clean pipeline. The architecture supports full temporal sequences for future integration with streaming weather feeds.

Categorical Identifiers County and state IDs (learned embeddings) and NLCD land cover classification codes, enabling the model to capture region-specific patterns.

3.2.2 Encoder Architecture

Static Encoder Linear projection from input dimension to 128-dimensional hidden space, padded to a fixed 50-dimensional input:

$$h_{\text{static}} = W \cdot \text{pad}(x_{\text{static}}, 50) \tag{1}$$

Transformer with Diffusion Attention Three transformer layers with 4-head diffusion (heat kernel) attention (detailed in Section 4). Each layer uses depth-scaled diffusion times, pre-norm architecture with LayerNorm, and GELU activation in the feed-forward network (intermediate dimension 512).

Per-Hazard LoRA Heads Each hazard type has a dedicated prediction head using Low-Rank Adaptation (rank 8) to specialize predictions while sharing the transformer backbone. This enables hazard-specific fine-tuning without proportionally increasing parameter count.

4 Methodological Context: Diffusion-Based Attention

4.1 Background

This project focuses on the applied evaluation of a diffusion-based attention mechanism within an emergency management decision-support system. The attention formulation used in HazardLM-Diffusion draws on prior theoretical and empirical work by the author, which established the mathematical foundations and validated the approach on language modeling benchmarks [1, 2].

4.2 Theoretical Foundation

Standard transformer attention computes weights using the softmax function:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{2}$$

As discussed in prior work [1], softmax attention can be understood as the equilibrium distribution of a drift-diffusion process. The attention weights represent the stationary solution to a Fokker-Planck equation where the drift is determined by query-key similarity.

Heat kernel attention generalizes this by stopping the diffusion process at finite time t:

$$\alpha_j^{(t)} = \frac{\exp(-\|q - k_j\|^2/4t)}{\sum_i \exp(-\|q - k_i\|^2/4t)} \tag{3}$$

For normalized query and key vectors, this is equivalent to softmax with temperature $T = 2t$, but the diffusion interpretation provides additional theoretical guarantees including provable locality properties and improved calibration.

4.3 Application in HazardLM-Diffusion

In the HazardLM-Diffusion architecture, diffusion attention replaces standard softmax attention in all three transformer layers. Each layer learns a depth-scaled diffusion time parameter, allowing different layers to operate at different temporal resolutions. The mechanism provides several design benefits:

- **Adaptive locality**: Shallow layers can use short diffusion times for local pattern detection while deeper layers use longer times for global context aggregation
- **Outlier robustness**: The Gaussian kernel form provides natural down-weighting of anomalous inputs compared to the exponential-dot-product form of softmax
- **Interpretable dynamics**: The learned diffusion time parameter provides a tunable "deliberation time" that balances responsiveness and stability

The theoretical motivation that different contexts benefit from different deliberation times [1] is validated by the observation that learned diffusion times evolve during training (Section 7.4).

5 Training Methodology

5.1 Dataset Construction

The clean labeled dataset spans 2000 to 2025, constructed through a rigorous pipeline described in Section 8:

Table 2: Training Dataset Statistics

Metric	**Value**
Total observations	370,383
Training split (<80% by date)	∼296,000
Validation split (80–90%)	∼37,000
Test split (>90%)	∼37,000
Counties	39
Date range	2000-01-01 to 2025-12-31

5.2 Feature Pipeline

The production dataset uses 21 curated features organized into three categories:

- **Geographic**: Latitude, longitude, elevation, forest fraction, urban fraction, population density
- **Temporal**: Day of year, month, year (enabling seasonal awareness)
- **Meteorological**: Maximum/minimum temperature (`tmmx`/`tmmn`), relative humidity range (`rmin`/`rmax`), wind speed (`vs`), Energy Release Component (`erc`), precipitation (`pr`), vapor pressure deficit (`vpd`), red flag warnings, and 3-day rolling means for temperature, precipitation, and wind speed

This represents a deliberate simplification from earlier iterations that used 142+ engineered features. The reduction was driven by the discovery of data leakage in the original pipeline (Section 8) and the finding that simpler features with strict geographic constraints produced more reliable labels and generalizable models.

5.3 Label Construction

Binary hazard labels are assigned using a 3-day event window: a county-day observation is labeled positive if a verified hazard event occurred within 3 days in that county. Labels are constructed from NOAA Storm Events (flood, wind, winter), WFIGS wildfire records (fire), and USGS earthquake catalog (seismic, $M \geq 3.0$). Strict county-level geographic matching ensures that events are only attributed to the county where they occurred.

Table 3: Hazard Label Base Rates

Hazard	**Base Rate**	**Data Source**
Fire	12.80%	WFIGS Wildfire Locations
Flood	1.27%	NOAA Storm Events
Wind	3.62%	NOAA Storm Events
Winter	3.62%	NOAA Storm Events
Seismic	2.87%	USGS Earthquake Catalog

5.4 Class Imbalance Handling

Hazard events are rare (1–13% positive rates). This class imbalance is addressed through focal loss with seasonal penalties:

$$\mathrm{FL}(p_t) = -\alpha_t (1 - p_t)^\gamma \log(p_t) \tag{4}$$

where $\gamma = 2.0$ down-weights easy negatives and $\alpha = 0.75$ balances class frequencies.

Additionally, seasonal penalty multipliers (3× loss weight) are applied to physically implausible false positives: fire predictions during November–March, winter storm predictions during June–September, and high-wind predictions during December–February. These penalties encode domain knowledge that wildfire in Washington's winter and winter storms in summer are climatologically near-impossible.

5.5 Optimization

The model is trained for 30 epochs using AdamW with weight decay 10^{-4}, base learning rate 3×10^{-4} with OneCycleLR scheduling (max LR 10^{-3}), gradient clipping with max norm 1.0, and batch size 256. The temporal train/validation/test split (by date, not random) ensures no future data leaks into training.

6 Post-Hoc Calibration Pipeline

Raw neural network outputs are often poorly calibrated—the model may output 50% probability for events that occur only 3% of the time. AHI addresses this with a multi-stage calibration pipeline applied at inference time.

6.1 Temperature Scaling

Per-hazard temperature parameters T_h are fitted on the validation set by minimizing negative log-likelihood:

$$T_h^* = \arg\min_{T>0} \sum_i -\left[y_i \log \sigma(z_i/T) + (1 - y_i) \log(1 - \sigma(z_i/T))\right] \tag{5}$$

where z_i is the raw logit and σ is the sigmoid function. Temperature scaling preserves the model's ranking (AUC is invariant to monotonic transformations) while adjusting probability magnitudes.

Table 4: Fitted Temperature Scales (Validation Set)

Hazard	**Temperature**	**Effect**
Fire	0.436	Sharpens (good discrimination)
Flood	0.272	Sharpens (reduces overconfidence)
Wind	1.000	Clamped (weak head, $T < 1$ amplifies errors)
Winter	0.660	Moderate sharpening
Seismic	1.000	Clamped (near-random head)

For hazard heads with test AUC below 0.65 (wind, seismic), the fitted temperature is clamped to $T \geq 1.0$ to prevent sharpening of unreliable predictions.

6.2 Seasonal Prior

A physics-informed logit bias is applied based on calendar month, encoding Washington State climatology. The bias operates in log-odds space before the sigmoid:

$$z_{\text{final}} = z_{\text{scaled}} + b_{\text{seasonal}}(\text{hazard}, \text{month}) \tag{6}$$

Key seasonal adjustments include:

- **Fire**: -3.0 logit bias in December–February (virtually no WA wildfire), 0.0 in June–September (fire season)
- **Winter**: -3.0 logit bias in June–August (no winter storms), 0.0 in November–March (peak season)
- **Wind**: Mild -0.3 to -0.5 bias in summer months
- **Flood, Seismic**: No seasonal bias (year-round / tectonic)

6.3 Base-Rate Recalibration and Ceilings

Two additional corrections address systematically overconfident prediction heads:

Weak-head bias. For hazard types where the model achieves near-random discrimination (wind AUC 0.59, seismic AUC 0.50), a constant negative logit bias pulls the mean prediction toward the historical base rate. Wind receives a -1.5 bias and seismic receives -2.5, preserving whatever ranking signal exists while anchoring to realistic probabilities.

Base-rate ceiling. A hard probability cap prevents physically implausible values:

Table 5: Base-Rate Ceilings by Hazard Type

Hazard	Ceiling	Rationale
Fire	35%	Well-calibrated head (AUC 0.73), generous ceiling
Flood	25%	Moderate head (AUC 0.65)
Wind	15%	Weak head (AUC 0.59), can't trust high values
Winter	35%	Best head (AUC 0.74), generous ceiling
Seismic	8%	Near-random head (AUC 0.50), tight ceiling

6.4 Calibration Pipeline Effect

The combined pipeline transforms raw model outputs into operationally plausible predictions. For Adams County in February:

Table 6: Calibration Pipeline Effect: Adams County, February

Hazard	Raw Prob	Calibrated	Primary Correction
Fire	49.2%	0.0%	Seasonal prior (-3.0 logit bias)
Flood	40.7%	0.0%	Temperature scaling ($T = 0.27$)
Wind	38.5%	0.1%	Weak-head bias (-1.5)
Winter	21.3%	0.0%	Temperature scaling
Seismic	17.3%	0.8%	Weak-head bias (-2.5) + ceiling

6.5 Operational Response Tiers

Calibrated probabilities map to operational response postures, enabling practitioners to select appropriate actions:

Table 7: Risk Level Thresholds and Response Postures

Probability	**Level**	**Recommended Response**
$< 15\%$	Low	Routine monitoring
15% to 30%	Moderate	Enhanced awareness
30% to 50%	Elevated	Active preparation
50% to 70%	High	Pre-position resources
$> 70\%$	Severe	Elevated readiness

These thresholds are calibrated against historical county-level data (2000 to 2025). Practitioners can adjust response postures based on local factors while using the probability as a defensible baseline.

7 Results and Validation

7.1 XGBoost Baseline

A per-hazard XGBoost classifier serves as the primary discrimination baseline, using the same 21 features with isotonic calibration:

Table 8: XGBoost Baseline: Test Set Performance

Hazard	**AUC**	**Avg. Precision**	**Brier Score**	**ECE**
Fire	0.870	0.815	0.141	0.054
Flood	0.714	0.049	0.009	0.008
Wind	0.713	0.055	0.026	0.004
Winter	0.885	0.233	0.034	0.016
Seismic	0.721	0.079	0.030	0.017
Mean	**0.781**	—	—	**0.020**

XGBoost achieves strong discrimination and excellent calibration (mean ECE 0.020) with isotonic post-processing, consistent with the literature showing gradient-boosted trees as strong baselines for tabular prediction tasks.

7.2 HazardLM-Diffusion Results

The diffusion attention model was trained for 30 epochs on the same dataset:

Table 9: HazardLM-Diffusion: Test Set Performance

Hazard	AUC	Avg. Precision	Raw ECE	Calibrated ECE
Fire	0.731	0.583	0.078	<0.05*
Flood	0.648	0.018	0.250	<0.08*
Wind	0.585	0.053	0.336	<0.15*
Winter	0.742	0.142	0.141	<0.06*
Seismic	0.499	0.031	0.308	<0.08*
Mean	**0.641**	—	**0.223**	—

*Calibrated ECE estimated from the post-hoc pipeline (temperature scaling + seasonal prior + base-rate corrections). Exact values depend on the evaluation period's seasonal composition.

7.3 Comparative Analysis

Table 10: Model Comparison: Test Set AUC by Hazard Type

Hazard	XGBoost	HazardLM-Diffusion	Gap
Fire	0.870	0.731	−0.139
Flood	0.714	0.648	−0.066
Wind	0.713	0.585	−0.128
Winter	0.885	0.742	−0.143
Seismic	0.721	0.499	−0.222
Mean	**0.781**	**0.641**	**−0.140**

The XGBoost baseline outperforms the diffusion model on all hazard types, with a mean AUC gap of 0.14. This is consistent with established findings that gradient-boosted trees remain highly competitive on structured tabular data [1]. However, the diffusion model provides several properties not available in tree-based models, including a unified architecture for all hazard types, learned attention patterns that can be inspected for interpretability, and a foundation for incorporating unstructured inputs (satellite imagery, text reports) in future work.

7.4 Training Dynamics

Training diagnostics revealed meaningful patterns in the adaptive diffusion time parameter. The mean diffusion time evolved from 0.34 at initialization, increasing as the model learned to utilize broader temporal context. Validation performance peaked around epoch 17–18 for most hazard types, with the final test evaluation conducted at epoch 30.

Training loss decreased from 0.54 to 0.22 over 30 epochs, indicating the model continued learning even after validation AUC plateaued—suggesting some overfitting in later epochs.

8 Data Pipeline Correction and Impact

8.1 Discovery of Data Leakage

During development, a comprehensive audit of the data pipeline revealed critical issues in the original label construction:

1. **Geographic constraint failure**: The labeling script's fallback logic dropped county-level matching, assigning state-wide events to individual counties. This inflated the positive label rate to 97.7%—rendering the model's apparent performance meaningless.
2. **Label window inflation**: A 30-day event window created massive false positives through temporal "halo effects." Switching to a 3-day window produced realistic base rates.
3. **Symptom**: The deployed dashboard showed Adams County at 49.9% fire risk in February—a physically implausible value that triggered the investigation.

8.2 Corrective Actions

The clean label pipeline (`build_clean_labels.py`) enforces:

- Strict county-level geographic matching with no fallback
- 3-day event window (reduced from 30 days)
- Verified source data from NOAA Storm Events, WFIGS, and USGS
- Temporal train/validation/test split (no random shuffle, preventing temporal leakage)

8.3 Impact on Reported Performance

The data correction is the primary driver of the difference between initial and final performance metrics:

Table 11: Performance Impact of Data Correction

Hazard	**Leaky Data AUC**	**Clean Data AUC**
Fire	0.72	0.73
Flood	0.63	0.65
Wind	0.48	0.59
Winter	0.65	0.74
Seismic	0.45	0.50
Mean	**0.585**	**0.641**

The clean data actually *improved* AUC for most hazard types despite producing more realistic (lower) base rates, because the model can now learn genuine hazard patterns rather than fitting to label noise. The corrected model produces more operationally accurate predictions, as confirmed by live testing against known historical events.

9 Operational Integration

9.1 User Interface Design

AHI provides multiple interfaces tailored to different users:

Interactive Dashboard The primary interface for daily operations includes real-time risk maps with color-coded hazard layers, click-to-expand county details with AI predictions, historical trend visualization, and climate-driven risk explanations. The dashboard integrates model evaluation tools showing training history, per-hazard AUC curves, and calibration diagnostics.

Automated Reporting Scheduled generation of daily situation awareness briefs, weekly trend analysis, monthly executive summaries, and FEMA grant application support documents.

9.2 Explainability for Non-Technical Users

Emergency managers need to understand *why* a prediction was made, not just what it is. AHI provides:

Plain-Language Explanations

> "Wildfire risk is elevated because the Energy Release Component (89) is in the top 5% of historical values, combined with 14 consecutive days of below-normal precipitation. Similar conditions preceded 73% of historical fire declarations in this region."

Feature Importance Visualization Bar charts showing which input features most influenced each prediction, enabling rapid assessment of whether the model's reasoning aligns with local knowledge.

Historical Analogues When current conditions match past hazard events, the system identifies similar historical cases.

10 Discussion

10.1 Calibration as a First-Class Requirement

AHI's primary contribution is not discrimination (ranking hazards correctly) but *calibration* (producing probabilities that mean what they say). A model with AUC 0.95 that systematically overestimates risk by 20% forces emergency managers into defensive postures. A model with AUC 0.73 whose predictions are calibrated through a transparent pipeline enables informed escalation decisions, because practitioners understand both what the model says and the basis for that statement.

The multi-stage calibration pipeline (Section 6) makes the calibration process explicit and auditable. Each stage has a clear physical or statistical justification: temperature scaling optimizes log-likelihood on held-out data, seasonal priors encode domain knowledge about Washington climatology, and base-rate ceilings prevent the model from exceeding physically plausible risk levels. This transparency is essential for practitioner trust.

10.2 Honest Assessment of Diffusion Attention

The diffusion attention mechanism does not outperform XGBoost on this tabular prediction task. This result is consistent with the broader machine learning literature, which consistently finds gradient-boosted trees competitive or superior on structured tabular data. However, several points merit discussion:

1. **The diffusion model was trained on 21 features with limited temporal structure.** The clean pipeline's zero-padded temporal input gives the transformer's attention mechanism little to attend over. With streaming weather data providing actual 14-day sequences, the attention mechanism's temporal modeling capabilities would be more fully utilized.
2. **The architecture is designed for extensibility.** Tree-based models cannot natively process satellite imagery, text reports, or sequential weather data. The transformer backbone can incorporate these additional modalities without architectural changes.
3. **Diffusion attention provides interpretable dynamics.** The learned diffusion time parameter reveals how the model balances local and global context—a property with no analogue in gradient-boosted trees.
4. **Prior work demonstrated stronger results on language tasks.** The ECE reduction of 6–46% compared to softmax attention observed in language modeling benchmarks [2] may require richer sequential structure to manifest in tabular hazard prediction.

10.3 Positioning Against Alternatives

AHI occupies a specific niche in the emergency management tool landscape:

Table 12: AHI Positioning vs. Alternatives

Capability	**Cat Models**	**Prep Tools**	**AHI**
Multi-hazard prediction	Some	No	Yes
Calibrated probabilities	Yes	No	Yes
Operational confidence tiers	Limited	No	Yes
County resolution	Varies	Yes	Yes
Affordable	No	Yes	Yes
Open/auditable	No	Some	Yes

10.4 Limitations

Several limitations merit acknowledgment:

- **Geographic Scope**: Current training data covers Washington State only; expansion requires additional data integration and regional calibration of seasonal priors
- **Temporal Resolution**: Daily predictions; sub-daily resolution would require different data infrastructure
- **Weak Hazard Heads**: Wind (AUC 0.59) and seismic (AUC 0.50) prediction heads perform near-random discrimination. The post-hoc calibration pipeline constrains their outputs but cannot create signal where the model has not learned it. Practitioners should treat wind and seismic predictions as base-rate estimates rather than condition-specific forecasts
- **Seismic Prediction**: Earthquakes are fundamentally stochastic with respect to weather and temporal features. The near-random AUC (0.50) confirms that the available features carry no useful seismic signal. The model captures only the geographic prior (some counties are more seismically active than others)
- **Local Verification**: County-aggregated data may not capture sub-county variation; practitioners should verify with local observations for elevated predictions
- **Diffusion vs. XGBoost Gap**: The transformer-based model underperforms XGBoost by a mean of 0.14 AUC on this tabular dataset, consistent with known limitations of neural approaches on structured data without rich sequential or multimodal inputs

10.5 Future Development

Planned enhancements include:

- Integration of real-time weather feeds (NWS/NOAA) to populate the temporal encoder with actual 14-day sequences, addressing the current zero-padding limitation
- Satellite imagery encoder (Sentinel-2, GOES-West) for vegetation condition and thermal anomaly detection, leveraging the transformer's multimodal capability
- Expansion to Pacific Northwest states (Oregon, Idaho) with region-specific seasonal priors and hierarchical calibration
- Softmax attention ablation study on the identical architecture to directly quantify the contribution of diffusion attention vs. standard attention in this domain
- Improved seismic modeling through integration of USGS fault proximity data and historical recurrence intervals as static features

11 Conclusion

Adaptive Hazard Intelligence demonstrates that calibrated, multi-hazard risk prediction is achievable on commodity hardware and can be deployed in a form accessible to county emergency managers.

The core contribution is *decision control*: rather than presenting hazard information that requires practitioners to guess at appropriate responses, AHI returns calibrated probabilities mapped to operational confidence tiers. Emergency managers select response postures based on quantified risk levels, retaining control over escalation decisions rather than reacting to opaque alerts.

Key outcomes:

- Multi-hazard prediction across five hazard types with honest performance reporting (XGBoost mean AUC 0.78, diffusion model mean AUC 0.64)
- A transparent, multi-stage calibration pipeline that brings raw model outputs into alignment with historical base rates
- Discovery and correction of data leakage in the original label pipeline, demonstrating the importance of rigorous data validation in applied ML
- Operational confidence tiers that map to ICS-compatible response postures
- Live deployment on Streamlit Cloud with full source code on GitHub

This project validates that diffusion-based attention mechanisms can be applied in an operational emergency management context, while honestly acknowledging that the current implementation does not outperform gradient-boosted tree baselines on structured tabular data. The framework for integrating federal hazard data, constructing calibrated predictions, and deploying accessible decision-support tools can be replicated and adapted for other states.

AHI provides emergency managers with what commercial catastrophe models and preparedness tools do not: calibrated risk estimates with a transparent methodology, accessible to jurisdictions regardless of budget.

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Code and Data Availability

All code is available at: https://github.com/JDCurry/ahi-capstone
Federal datasets are publicly available through their respective agencies.
Processed datasets available upon request for research purposes.

References

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